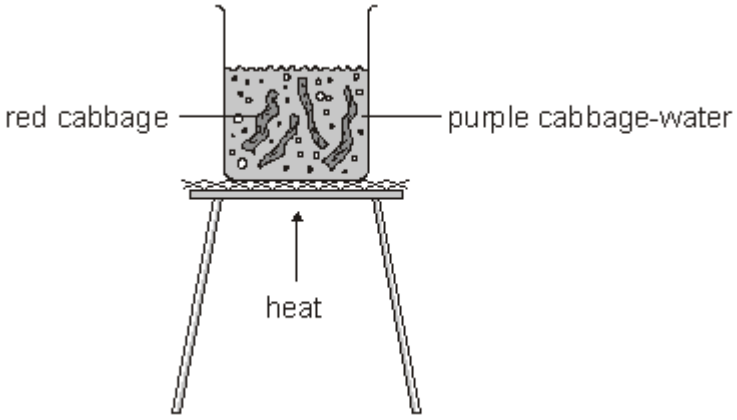
Q1.

Sharna boiled some red cabbage in water. The cabbage-water turned purple.



(a) (i) Sharna separated pieces of cabbage from the cabbage-water.

Which method did she use?
Tick the correct box.

chromatography	☐	filtration	☐
condensation	☐	freezing	☐

1 mark

(ii) Sharna wanted to find out if the purple cabbage-water contained more than one **coloured** substance.

Which method did she use?
Tick the correct box.

chromatography	☐	filtration	☐
condensation	☐	freezing	☐

1 mark

(b) Sharna mixed the purple cabbage-water with some other liquids.
She wrote the colours of the mixtures in a table as shown below.

	colour of cabbage-water mixed with liquid	**Is the liquid acidic, alkaline or neutral?**
liquid 1	red	acidic
liquid 2	blue	alkaline
liquid 3	purple	neutral

Use the information in the table to answer parts (i) and (ii) below.

(i) Sharna mixed cabbage-water with colourless washing-up liquid.
The mixture turned **blue.**

What does this tell you about the washing-up liquid?

.....

1 mark

(ii) Sharna then mixed cabbage-water with lemon juice.
Lemon juice is **acidic.**

What colour was the mixture?

.....

1 mark

(c) What is the name of a chemical which changes colour when it is mixed with acids or alkalis?
Tick the correct box.

filtrate	☐	indicator	☐
non-metal	☐	solution	☐

1 mark
maximum 5 marks

Q1.

(a) (i) filtration ✓

if more than one box is ticked, award no mark

1 (L3)

(ii) chromatography ✓

if more than one box is ticked, award no mark

1 (L3)

(b) (i) it is alkaline

accept 'alkali'

1 (L3)

(ii) red

1 (L3)

(c) indicator ✓

if more than one box is ticked, award no mark

1 (L4)

[5]